Lecture 1: Foundations of Molecular Dynamics

1. Why Molecular Dynamics?

Molecular dynamics (MD) provides a computational framework for following the motion of atoms and molecules in time. At its core, MD solves Newton's equations of motion,

$$M_I \frac{d^2\mathbf{R}_I}{dt^2} = \mathbf{F}_I(\mathbf{R}), \tag{1}$$

where M_I and $\mathbf{R}_I$ are the mass and position of nucleus I, and the force is obtained from a potential-energy function,

$$\mathbf{F}_I(\mathbf{R}) = -\nabla_I U(\mathbf{R}). \tag{2}$$

A molecular dynamics trajectory therefore consists of a sequence of molecular configurations,

$$\mathbf{R}(0), \mathbf{R}(\Delta t), \mathbf{R}(2\Delta t), \ldots, \tag{3}$$

together with the corresponding velocities.

The usefulness of MD extends beyond visualization of molecular motion. Under appropriate conditions, an MD trajectory provides samples from a statistical mechanical ensemble and can therefore be used to calculate equilibrium properties.

2. From Molecular Quantum Mechanics to Classical Dynamics

A molecular system is fundamentally quantum mechanical. The full molecular wavefunction depends on both electronic and nuclear coordinates,

$$\Psi(\mathbf{r}, \mathbf{R}, t), \tag{4}$$

and obeys the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi. \tag{5}$$

The molecular Hamiltonian can be separated schematically into

$$\hat{H} = \hat{T}_N + \hat{H}_e(\mathbf{r}; \mathbf{R}), \tag{6}$$

where $\hat{T}_N$ is the nuclear kinetic-energy operator and $\hat{H}_e$ contains the electronic kinetic energy and the Coulomb interactions.

Because nuclei are much heavier than electrons, electronic motion typically occurs on a much faster timescale than nuclear motion. This motivates the Born–Oppenheimer approximation.

For a fixed nuclear configuration $\mathbf{R}$, one solves the electronic Schrödinger equation,

$$\hat{H}_e(\mathbf{r}; \mathbf{R})\phi_k(\mathbf{r}; \mathbf{R}) = E_k(\mathbf{R})\phi_k(\mathbf{r}; \mathbf{R}). \tag{7}$$

The nuclear coordinates appear as parameters in this equation. For ground-state dynamics, the electronic ground-state energy

$$E_0(\mathbf{R}) \tag{8}$$

defines a potential-energy surface over nuclear configuration space.

Neglecting nonadiabatic transitions, the nuclear wavefunction evolves according to

$$i\hbar\frac{\partial\chi(\mathbf{R},t)}{\partial t} = \left[-\sum_I \frac{\hbar^2}{2M_I}\nabla_I^2 + E_0(\mathbf{R})\right]\chi(\mathbf{R},t). \tag{9}$$

The Born–Oppenheimer approximation therefore does not itself make the nuclei classical. It separates fast electronic motion from slow nuclear motion and provides the potential-energy surface on which the nuclei move.

3. The Classical Nuclear Approximation

For a single coordinate R, Ehrenfest's theorem gives

$$M\frac{d^2}{dt^2}\langle\hat{R}\rangle = -\left\langle\frac{dE_0(\hat{R})}{dR}\right\rangle. \tag{10}$$

In general,

$$\left\langle E_0'(\hat{R})\right\rangle \neq E_0'\left(\langle\hat{R}\rangle\right). \tag{11}$$

If the nuclear wavepacket is sufficiently localized, however,

$$\left\langle E_0'(\hat{R})\right\rangle \approx E_0'\left(\langle\hat{R}\rangle\right), \tag{12}$$

and the center of the nuclear wavepacket approximately follows

$$M\ddot{R} = -\frac{dE_0(R)}{dR}. \tag{13}$$

For many nuclei,

$$\boxed{M_I\ddot{\mathbf{R}}_I = -\nabla_I E_0(\mathbf{R})} \tag{14}$$

Thus the conceptual sequence is

$$\boxed{\text{molecular quantum mechanics} \to \text{Born–Oppenheimer potential-energy surface} \to \text{classical nuclear dynamics}}. \tag{15}$$

4. Where Does the Potential Energy Come From?

The equations of motion require a potential-energy function

$$U(\mathbf{R}). \tag{16}$$

Different molecular simulation methods differ mainly in how this function is obtained.

Ab initio molecular dynamics. The electronic structure problem is solved repeatedly during the simulation,

$$U(\mathbf{R}) = E_{\text{electronic}}(\mathbf{R}). \tag{17}$$

The forces are calculated as

$$\mathbf{F}_I = -\nabla_I E_{\text{electronic}}(\mathbf{R}). \tag{18}$$

Classical force fields. The electronic potential-energy surface is approximated by an analytical function,

$$U_{\mathrm{FF}} = U_{\mathrm{bond}} + U_{\mathrm{angle}} + U_{\mathrm{dihedral}} + U_{\mathrm{nonbonded}}. \tag{19}$$

A common nonbonded form is

$$U_{\mathrm{nonbonded}} = \sum_{i<j} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right] + \sum_{i<j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}. \tag{20}$$

Machine-learning force fields. Modern models replace fixed analytical functional forms by flexible parameterized functions,

$$U(\mathbf{R}) \approx U_\theta(\mathbf{R}), \tag{21}$$

where θ denotes parameters learned from quantum calculations, experimental data, or statistical ensembles.

Regardless of the representation of U, the molecular dynamics equation remains

$$M_I \ddot{\mathbf{R}}_I = -\nabla_I U(\mathbf{R}). \tag{22}$$

5. Numerical Integration of Newton's Equations

The equations of motion cannot usually be solved analytically for a many-body molecular system. Instead, they are integrated numerically using a finite time step Δt.

Expanding the position in a Taylor series,

$$\mathbf{R}(t + \Delta t) = \mathbf{R}(t) + \mathbf{V}(t)\Delta t + \frac{1}{2}\mathbf{A}(t)\Delta t^2 + \mathcal{O}(\Delta t^3). \tag{23}$$

A widely used algorithm is velocity Verlet:

$$\mathbf{R}(t + \Delta t) = \mathbf{R}(t) + \mathbf{V}(t)\Delta t + \frac{1}{2}\mathbf{A}(t)\Delta t^2, \tag{24}$$

$$\mathbf{V}(t + \Delta t) = \mathbf{V}(t) + \frac{1}{2}\left[\mathbf{A}(t) + \mathbf{A}(t + \Delta t)\right]\Delta t. \tag{25}$$

The acceleration is determined from

$$\mathbf{A}_I = \frac{\mathbf{F}_I}{M_I}. \tag{26}$$

The time step must be sufficiently short to resolve the fastest relevant molecular motions. In atomistic biomolecular simulations, bond vibrations involving hydrogen often impose a timestep on the order of femtoseconds.

The essential MD loop is therefore

$$\boxed{\mathbf{R} \to U(\mathbf{R}) \to \mathbf{F}(\mathbf{R}) \to \mathbf{R}(t + \Delta t)}. \tag{27}$$

- The update scheme for velocity ensures that the expression for positions is accurate up $\mathcal{O}(\Delta t^4)$.
- Velocity verlet is time reversible. Time reversible symmetry is an important feature of Hamiltonian dynamics.
- Velocity verlet is a symplectic algorithm and does not lead to long time drifts in energy.

6. Conservation of Energy and the Microcanonical Ensemble

For an isolated classical system with Hamiltonian

$$H(\mathbf{R},\mathbf{P}) = \sum_I \frac{\mathbf{P}_I^2}{2M_I} + U(\mathbf{R}), \tag{28}$$

Hamiltonian dynamics conserves the total energy,

$$E = K + U. \tag{29}$$

Therefore an ideal Newtonian MD trajectory evolves at fixed

$$N, \quad V, \quad E, \tag{30}$$

corresponding to the microcanonical or NVE ensemble.

The microcanonical probability distribution may be written as

$$\rho_{\mathrm{NVE}}(\mathbf{R},\mathbf{P}) \propto \delta\left[H(\mathbf{R},\mathbf{P}) - E\right]. \tag{31}$$

Thus molecular dynamics is not simply a deterministic trajectory-generation procedure. It can also serve as a statistical sampling method.

7. Time Averages and Ensemble Averages

Suppose we want the equilibrium average of an observable A.

Statistical mechanics defines the ensemble average as

$$\langle A \rangle = \int d\mathbf{R}\, d\mathbf{P}\, \rho(\mathbf{R},\mathbf{P}) A(\mathbf{R},\mathbf{P}). \tag{32}$$

An MD simulation instead produces a time series,

$$A(t). \tag{33}$$

The corresponding time average is

$$\overline{A} = \lim_{\tau\to\infty} \frac{1}{\tau} \int_0^\tau A(t)\, dt. \tag{34}$$

If the dynamics is ergodic,

$$\boxed{\overline{A} = \langle A \rangle}. \tag{35}$$

This assumption provides the connection between a molecular trajectory and equilibrium statistical mechanics.

In practice, ergodicity may fail or convergence may be extremely slow when large free-energy barriers separate important regions of configuration space. This limitation motivates enhanced-sampling methods.

8. The Canonical Ensemble

Many experiments occur at approximately constant temperature rather than constant total energy.

In the canonical ensemble,

$$\rho(\mathbf{R}, \mathbf{P}) = \frac{1}{Z} \exp\left[-\beta H(\mathbf{R}, \mathbf{P})\right], \tag{36}$$

where

$$\beta = \frac{1}{k_{\mathrm{B}} T}. \tag{37}$$

The configurational distribution is

$$p(\mathbf{R}) = \frac{1}{Z_{\mathrm{conf}}} e^{-\beta U(\mathbf{R})}. \tag{38}$$

This equation is one of the central equations of molecular simulation:

$$\boxed{p(\mathbf{R}) \propto e^{-\beta U(\mathbf{R})}}. \tag{39}$$

It establishes a direct relationship between energy and probability.

Low-energy configurations are more probable, while high-energy configurations are exponentially suppressed.

9. Langevin Dynamics and Constant-Temperature Sampling

A simple way to generate a canonical ensemble is to couple the system to a thermal bath.

For a single coordinate,

$$m\ddot{x} = -\frac{dU}{dx} - \gamma m \dot{x} + R(t). \tag{40}$$

The three terms correspond to

1. the conservative molecular force,
2. friction from the environment,
3. random collisions with the environment.

The random force satisfies

$$\langle R(t) \rangle = 0 \tag{41}$$

and

$$\langle R(t) R(t') \rangle = 2\gamma m k_{\mathrm{B}} T \delta(t - t'). \tag{42}$$

The magnitude of the random force and the friction coefficient are therefore not independent. Their relationship is an example of the fluctuation–dissipation theorem.

The stationary distribution generated by Langevin dynamics is the canonical distribution,

$$\rho(x, p) \propto \exp\left[-\beta \left(\frac{p^2}{2m} + U(x)\right)\right]. \tag{43}$$

Thus Langevin dynamics provides both a dynamical model and a sampling algorithm for the NVT ensemble.